

The main question of this talk was: Where are all the black holes in the Milky Way?

Scientists estimate that our galaxy should contain around 100 million stellar mass black holes, because many massive stars become black holes after they die. However, astronomers have discovered only about 100 blackhole candidates, and only around 30 have well measured masses. In the past, astronomers usually found black holes in x ray binaries. In these systems, a black hole pulls material from a nearby star. The falling material becomes very hot and produces x rays. However, most black holes do not actively take material from another star, so they are dark and difficult to find.

The Gaia spacecraft gives astronomers a new way to discover these dormant black holes. Gaia measures the positions and movements of stars extremely precisely. If a visible star moves around an invisible but massive object, astronomers can study its orbit and use Kepler's laws to estimate the mass of the invisible companion. They can also measure the star's radial velocity to understand its complete three dimensional motion.

Gaia has helped scientists discover three important systems: 1. Gaia BH1 contains a sun like star orbiting a black hole of about nine solar masses. 2. Gaia BH2 contains an evolved star orbiting another black hole of about nine solar masses. 3. Gaia BH3 contains a black hole of about 33 solar masses and a very metal poor companion star.

Gaia BH3 suggests that metallicity may affect black hole mass. Metal poor stars have weaker stellar winds, so they lose less material before dying. As a result, they may produce more massive black holes. However, metallicity is not the only factor; the star's original mass, rotation, companions, and environment can also matter. Another important question is how black hole binaries form. A massive star expands into a red supergiant before becoming a black hole, so it could swallow a nearby low mass companion. Possible explanations include common envelope evolution, triple star systems, or interactions between stars inside a cluster.

I also learned that black holes may receive a natal kick when they form. Some black holes appear to receive strong kicks, while the black hole in the V404 Cygni triple system must have received a very weak kick. This suggests that black holes can form in different ways.

Finally, future Gaia data should reveal many more dormant black holes. The Roman Space Telescope may also find isolated black holes through gravitational microlensing. Overall, this talk showed me that astronomers can discover invisible objects by carefully measuring how their gravity affects visible stars.